

OPTICAL MUSIC RECOGNITION FOR JAZZ LEAD SHEETS

A CRNN–CTC approach for monophonic lines

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Resum (Català)

Paraules clau: Optical Music Recognition, CRNN, CTC, LMX, Real Book.

[TODO: 1–2 pages max. Summary in Catalan.]

Resumen (Español)

Palabras clave: Optical Music Recognition, CRNN, CTC, LMX, Real Book.

[TODO: 1–2 pages max. Summary in Spanish.]

Abstract (English)

Keywords: Optical Music Recognition, CRNN, CTC, LMX, Real Book.

[TODO: 1–2 pages max. Summary in English.]

Contents

Resum (Català)	i
Resumen (Español)	ii
Abstract (English)	iii
1 Introduction and Context	1
1.1 Motivation	1
1.2 Problem Formulation	1
1.3 Objectives	1
1.4 Scope	1
2 State of the Art	2
2.1 Historical Approaches	2
2.2 Deep Learning Paradigm: CRNN and CTC	2
2.3 Modern End-to-End Architectures	2
2.4 Output Representations	2
3 Project Management (GEP Integration)	3
3.1 Justification of Alternatives	3
3.2 Project Planning	3
3.3 Economic Management	3
4 Methodology and Architecture	4
4.1 The Data Pipeline	4
4.1.1 Synthetic generation with LilyPond and LilyJAZZ	4
4.1.2 Data cleaning	4
4.1.3 Noise engine	4
4.2 Neural Network Architecture	4
4.2.1 ResNet-18 backbone	4
4.2.2 Asymmetric strides and temporal resolution	4
4.3 Training Loop	4
4.4 Bridging the Domain Gap (Fine-Tuning)	4
5 Results and Evaluation	5
5.1 Synthetic Baseline Results	5
5.2 The Domain Gap (Zero-Shot Testing)	5
5.3 Out-Of-Vocabulary (OOV) Solutions	5
5.4 Fine-Tuning Results	5

6	Sustainability and Ethical Implications	6
6.1	Environmental Sustainability	6
6.2	Economic Sustainability	6
6.3	Social Sustainability & Ethics	6
7	Conclusions and Future Work	7
7.1	Conclusions	7
7.2	Future Lines of Research	7
A	Additional Tables and Hyperparameters	8
B	Qualitative Predictions	9
C	Code Snippets	10

List of Figures

List of Tables

Chapter 1

Introduction and Context

1.1 Motivation

[TODO: Preservation of musical heritage, digital musicology, accessibility, education.]

1.2 Problem Formulation

[TODO: Challenge of transcribing sheet music; focus on handwritten/complex lead sheets; domain gap.]

1.3 Objectives

- *[TODO: Develop a CRNN-CTC pipeline for monophonic staff-line OMR.]*
- *[TODO: Achieve low Symbol Error Rate (SER) on synthetic validation/test sets.]*
- *[TODO: Handle out-of-vocabulary symbols via normalisation / mapping.]*

1.4 Scope

Included:

- *[TODO: Monophonic lines; synthetic data generation; augmentation; fine-tuning to real scans.]*

Excluded:

- *[TODO: Polyphony; full-page reading order / layout analysis beyond line-level.]*

Chapter 2

State of the Art

2.1 Historical Approaches

[TODO: Traditional heuristic/rule-based pipelines; staff detection/removal; why OpenCV morphology was insufficient.]

2.2 Deep Learning Paradigm: CRNN and CTC

[TODO: CRNN as visual encoder + sequence model; CTC alignment; why it fits left-to-right monophonic lines.]

2.3 Modern End-to-End Architectures

[TODO: Transformer-based OMR (Sheet Music Transformer, LEGATO, etc.) and trade-offs.]

2.4 Output Representations

*[TODO: LMX, **kern, agnostic vs. semantic; why LMX linearisation is used here.]*

Chapter 3

Project Management (GEP Integration)

3.1 Justification of Alternatives

[TODO: CRNN-CTC vs. classic CV vs. large transformers; constraints and rationale.]

3.2 Project Planning

[TODO: Gantt, milestones, deviations/risks; link to GEP deliverables.]

3.3 Economic Management

[TODO: Budget: hardware, hours costed as junior engineer, licences (if any), contingency.]

Chapter 4

Methodology and Architecture

4.1 The Data Pipeline

4.1.1 Synthetic generation with LilyPond and LilyJAZZ

[TODO: Rendering pipeline; notation constraints; dataset size targets.]

4.1.2 Data cleaning

[TODO: Filtering degenerate samples; criteria; impact (e.g., 26k removed).]

4.1.3 Noise engine

[TODO: Perspective warp, blur, scan-like artefacts; calibration vs. Real Book scans.]

4.2 Neural Network Architecture

4.2.1 ResNet-18 backbone

[TODO: Custom ResNet-18; feature map shapes; why chosen.]

4.2.2 Asymmetric strides and temporal resolution

[TODO: Height-to-1 squashing while preserving width for CTC time steps; math/intuition.]

4.3 Training Loop

[TODO: PyTorch AMP, OneCycleLR, early stopping, checkpoints, config choices.]

4.4 Bridging the Domain Gap (Fine-Tuning)

[TODO: Small Real Book dataset; freezing early layers; transfer learning strategy.]

Chapter 5

Results and Evaluation

5.1 Synthetic Baseline Results

[TODO: SER improvements (e.g., 7.4% $\rightarrow$ 1.17%); tables/plots; ablations.]

5.2 The Domain Gap (Zero-Shot Testing)

[TODO: Performance on real handwritten scan pre-fine-tuning; qualitative errors.]

5.3 Out-Of-Vocabulary (OOV) Solutions

[TODO: Pipeline updates; normalisation strategy (e.g., repeat signs $\rightarrow$ barlines).]

5.4 Fine-Tuning Results

[TODO: Post fine-tuning evaluation; Autumn Leaves read without text hallucinations.]

Chapter 6

Sustainability and Ethical Implications

6.1 Environmental Sustainability

[TODO: Training energy, GPU hours, carbon footprint; mitigations: AMP, early stopping.]

6.2 Economic Sustainability

[TODO: Cost/benefit vs commercial tools and manual transcription.]

6.3 Social Sustainability & Ethics

[TODO: Open-sourcing, accessibility, education; data/licensing considerations.]

Chapter 7

Conclusions and Future Work

7.1 Conclusions

[TODO: Key achievements; framing as Seq2Seq; synthetic SER; domain-gap fine-tuning.]

7.2 Future Lines of Research

[TODO: Full-page transcription (layout analysis / YOLO), polyphony, broader vocab, transformers.]

Appendix A

Additional Tables and Hyperparameters

[TODO: hyperparameter sweeps, training configs, etc.]

Appendix B

Qualitative Predictions

[TODO: extra figures/predictions that would clutter the main text.]

Appendix C

Code Snippets

[TODO: only the most relevant snippets; keep main text clean.]